

HILL CIPHER

1. INTRODUCTION:

Hill Cipher is a classical encryption technique based on linear algebra. It was developed by Lester S. Hill in 1929. It uses matrix multiplication to encrypt groups of letters in a message. Unlike simple substitution ciphers, Hill Cipher works on blocks of letters, making it more secure. Each letter is converted into a number, and encryption is performed using a key matrix. It is an important concept in modern cryptography studies.

2. CONCEPT:

- Hill Cipher is a polygraphic substitution cipher
- Developed by Lester S. Hill in 1929
- Uses matrix multiplication for encryption
- Works on blocks of letters instead of single letters
- Letters are converted into numbers (A=0, B=1, ..., Z=25)
- Uses a key matrix for encryption and decryption
- Requires inverse matrix for decryption

3. FORMULA:

- Encryption Formula: $C = (K \times P) \bmod 26$
 - $C \rightarrow$ Ciphertext vector
 - $K \rightarrow$ Key matrix ($n \times n$)
 - $P \rightarrow$ Plaintext vector ($n \times 1$)
 - $\bmod 26 \rightarrow$ Because we use 26 letters (A–Z)
- Decryption Formula: $P = (K^{-1} \times C) \bmod 26$
 - $K^{-1} \rightarrow$ Inverse of key matrix ($\bmod 26$)

4.IMPLEMENTATION:

1. Choose a key matrix (n x n) where determinant is non-zero
2. Convert plaintext into numerical values
3. Divide message into blocks of size n
4. Multiply each block with key matrix
5. Take modulo 26 of the result
6. Convert numbers back to letters to get ciphertext
7. For decryption, use inverse of key matrix

4.EXAMPLE:

Hill cipher:

$$C = KP \text{ mod } 26$$

$$P = K^{-1}C \text{ mod } 26$$

$$K = \begin{vmatrix} 9 & 4 \\ 5 & 7 \end{vmatrix} \quad PT = \begin{matrix} 22 & 12 \\ 4 & 22 \end{matrix}$$

$$C = \begin{vmatrix} 9 & 4 \\ 5 & 7 \end{vmatrix} \begin{vmatrix} 22 \\ 4 \end{vmatrix} \text{ mod } 26$$

$$= \begin{vmatrix} 198 + 16 \\ 110 + 28 \end{vmatrix} \text{ mod } 26$$

$$= \begin{vmatrix} 214 \\ 138 \end{vmatrix} \text{ mod } 26$$

$$= \begin{vmatrix} 6 \\ 8 \end{vmatrix} \rightarrow \begin{matrix} G \\ I \end{matrix}$$

$$P = K^{-1}C \text{ mod } 26$$

$$K^{-1} = d^{-1} \text{adj}(K) \text{ mod } 26$$

$$d^{-1} = d \times x = 1 \text{ mod } 26$$

$$43 \times 49 = 1 \text{ mod } 26$$

$$d = \begin{vmatrix} 9 & 4 \\ 5 & 7 \end{vmatrix} = d = (9 \times 7) - (5 \times 4) \\ = 63 - 20 = 43$$

$$\text{adj}(K) = \begin{vmatrix} 49 & -4 \\ -5 & 9 \end{vmatrix} \text{ mod } 26$$

$$= \begin{vmatrix} 343 & -196 \\ -245 & 441 \end{vmatrix} \text{ mod } 26$$

$$= \begin{vmatrix} 5 & -14 \\ -11 & 25 \end{vmatrix} \text{ mod } 26 \quad (\text{instead of } -11 \text{ symbol } +26)$$

$$K^{-1} = \begin{vmatrix} 5 & 12 \\ 15 & 25 \end{vmatrix} \begin{vmatrix} 6 \\ 8 \end{vmatrix} \text{ mod } 26$$

$$= \begin{vmatrix} 126 \\ 290 \end{vmatrix} \text{ mod } 26 \quad \begin{matrix} 30 + 96 \\ 90 + 200 \end{matrix}$$

$$= \begin{vmatrix} 22 \\ 4 \end{vmatrix} \rightarrow \begin{matrix} N \\ E \end{matrix}$$

5. REAL TIME EXPLANATION:

- **Secure Message Sharing (Basic Level)**
Two people can use a secret matrix key to send hidden messages so others cannot understand them.
- **Educational Purpose**
Widely used in cryptography classes to understand how matrix-based encryption works.
- **Foundation for Modern Cryptography**
Concepts like matrix operations and block encryption are used in advanced algorithms today.
- **Data Encoding Techniques**
Helps in understanding how data can be transformed and protected during transmission.
- **Research and Algorithm Design**
Used in developing new encryption ideas and studying weaknesses in classical ciphers.